

Sri Sivasubramaniya Nadar College of Engineering

Department of Information Technology

Course Code	UIT3361	Course Name	OBJECT-ORIENTED PROGRAMMING USING JAVA
Regulation	2024	Academic Year	2026-2027
Degree and Branch		B.Tech Information Technology	
Semester		III	
ASSIGNMENT			

Instructions to Students

Students are required to develop a standalone Java application based on the problem statement assigned to their batch. The application should demonstrate the concepts covered in the syllabus, including Core Java, Object-Oriented Programming, GUI programming using AWT/Swing/JavaFX, event handling, exception handling, and JDBC connectivity.

Student batches and their assigned problem statements are provided in the accompanying table.

As part of this exercise:

- The application must be implemented using Java and an appropriate GUI framework such as AWT, Swing, or JavaFX, as applicable to the assigned problem.
- The application should demonstrate appropriate object-oriented programming concepts, GUI components, layouts, event handling, input validation, and exception handling.
- JDBC connectivity must be implemented to connect the Java application with a relational database. The application should demonstrate basic database operations such as inserting, retrieving, updating, and deleting records.
- The database component is intended only to demonstrate JDBC connectivity and basic database operations. Students are not required to implement complex database design or ORM frameworks.
- Students should organize the application into appropriate classes and modules and follow good programming practices.
- Students are encouraged to apply suitable Design Patterns, including Creational, Structural, or Behavioral patterns, wherever appropriate.

Demonstration and Submission

- A live demonstration of the working application must be presented before the Model Examination.
- A comprehensive project report should be submitted along with the application. The report must include:
 - 1.System Requirements - Hardware and software specifications.
 - 2.System Design - Overall application architecture and GUI/database interaction.
 - 3.Core Modules Implemented - Description of the major modules and their functionality.
 - 4.JDBC Connectivity - Description of database connection and basic database operations performed.
 - 5.Code Snippets - Important code segments highlighting key functionality.
 - 6.Screenshots - Screenshots demonstrating the working application and its major features.
 - 7.Test Cases and Results - Test cases used to verify the functionality of the application.
 - 8.GitHub Repository Link - Link to the repository containing the complete project source code.

Students should ensure that the submitted application is functional, properly organized, and demonstrable, and that they are able to explain the Java concepts and implementation choices used in their application during the evaluation.

DEADLINE FOR SUBMISSION IN LMS - Oct 8,2026

S.No.	Problem Statement	Register Numbers
1	Student Information Management System Develop a Student Information Management System using Java and an appropriate GUI framework. The application should allow the user to enter, store, search, view, update, and delete student information through a graphical interface. The application should provide suitable GUI components for data entry and display, implement appropriate event handling and input validation, and use JDBC to connect the application to a relational database for storing and retrieving student records.	1 to 4
2	Employee Information Management System Develop an Employee Information Management System using Java and an appropriate GUI framework. The application should allow the user to add, view, search, modify, and delete employee information through the	5 to 8

	graphical interface. Appropriate GUI controls and event handling mechanisms should be used to perform the required operations. JDBC should be used to establish connectivity with a relational database and perform basic insert, retrieve, update, and delete operations on employee records.	
3	Library Book Management System Develop a Library Book Management System using Java and an appropriate GUI framework. The application should allow the user to add new books, view available books, search for a particular book, update book details, and remove book records. The system should provide a user-friendly graphical interface with appropriate event handling and input validation. JDBC connectivity should be implemented to store and retrieve the book information from a relational database.	9 to 12
4	Product Management System Develop a Product Management System using Java and an appropriate GUI framework. The application should allow the user to enter product details, view available products, search for a product, modify product information, and delete product records. The application should demonstrate the use of Java classes and objects, GUI components, event handling, and exception handling. JDBC should be used to establish database connectivity and perform basic CRUD operations on product records.	13 to 16
5	Personal Expense Management System Develop a Personal Expense Management System using Java and an appropriate GUI framework. The application should allow users to enter their expenses, categorize them, view previously entered expenses, search for specific expenses, modify expense details, and delete records. The application may also calculate and display the total expenditure. JDBC should be used to connect the Java application with a relational database and maintain the expense records.	17 to 20
6	Hospital Patient Information System Develop a Hospital Patient Information System using Java and an appropriate GUI framework. The application should allow the user to register patients, view patient information, search for a patient, update patient details, and delete patient records. The graphical interface should use suitable components and event handling techniques to provide an interactive application. JDBC connectivity should be implemented to store and retrieve patient information from a relational database.	21 to 24

7	Movie Ticket Booking System Develop a Movie Ticket Booking System using Java and an appropriate GUI framework. The application should allow users to view available movies and shows, select a show, enter booking details, view existing bookings, modify booking information, and cancel a booking. The application should demonstrate GUI components, event handling, input validation, and appropriate Java programming concepts. JDBC should be used to store and retrieve movie and booking information from a relational database.	25 to 28
8	Hotel Room Reservation System Develop a Hotel Room Reservation System using Java and an appropriate GUI framework. The application should allow users to view available rooms, enter guest details, make a room reservation, search for an existing reservation, modify reservation details, and cancel a reservation. The application should provide a suitable graphical interface with appropriate event handling and validation. JDBC should be used to connect the application with a relational database for storing and retrieving reservation information.	29 to 32
9	Vehicle Service Management System Develop a Vehicle Service Management System using Java and an appropriate GUI framework. The application should allow the user to register customer and vehicle details, create service records, search for vehicles, view service history, update service information, and delete records when required. The application should demonstrate Java OOP concepts, GUI programming, event handling, and exception handling. JDBC should be used for storing and retrieving vehicle and service information from a relational database.	33 to 36
10	Restaurant Table Reservation System Develop a Restaurant Table Reservation System using Java and an appropriate GUI framework. The application should allow users to enter customer details, view available tables, make reservations, search for existing reservations, modify reservation details, and cancel reservations. The application should provide an interactive graphical interface and implement suitable event handling and validation. JDBC should be used to maintain reservation information in a relational database.	37 to 40

11	Student Attendance Management System Develop a Student Attendance Management System using Java and an appropriate GUI framework. The application should allow the user to register students, record attendance, view attendance records, search for attendance information, modify attendance details, and delete records when necessary. The application may also calculate and display the attendance percentage of a student. JDBC should be used to store and retrieve student and attendance information from a relational database.	41 to 44
12	Employee Leave Management System Develop an Employee Leave Management System using Java and an appropriate GUI framework. The application should allow the user to enter employee details, submit leave requests, view leave records, search for leave requests, update the status of a request, and delete or cancel a leave request. The application should demonstrate appropriate Java programming, GUI components, event handling, and exception handling. JDBC should be used to maintain employee and leave information in a relational database.	45 to 48
13	Course Registration System Develop a Course Registration System using Java and an appropriate GUI framework. The application should allow students to view available courses, search for courses, register for a course, view registered courses, modify registration details, and cancel a registration. The application should use suitable GUI components and event handling mechanisms. JDBC should be used to store and retrieve student and course registration information from a relational database.	49 to 52
14	Online Quiz Management System Develop an Online Quiz Management System using Java and an appropriate GUI framework. The application should allow the user to add, view, search, modify, and delete quiz questions and enable students to attend the quiz through the graphical interface. The application should evaluate the answers and display the final score. JDBC should be used to store and retrieve questions and quiz-related information from a relational database.	53 to 56

15	Event Registration System Develop an Event Registration System using Java and an appropriate GUI framework. The application should allow the user to create and view event details, search for events, register participants, view participant information, modify registration details, and cancel registrations. The application should demonstrate Java OOP concepts, GUI programming, event handling, and input validation. JDBC should be used to store and retrieve event and participant information from a relational database.	57 to 60
16	Car Parking Management System Develop a Car Parking Management System using Java and an appropriate GUI framework. The application should allow the user to register vehicles, view available parking slots, record vehicle entry and exit, search for vehicle information, update parking details, and remove parking records. The application may also calculate parking charges based on the duration of parking. JDBC should be used to store and retrieve vehicle and parking information from a relational database.	61 to 64
17	Complaint Management System Develop a Complaint Management System using Java and an appropriate GUI framework. The application should allow users to register complaints, view submitted complaints, search for a complaint, update its status, and delete or close complaint records. The application should provide an interactive graphical interface with suitable event handling and input validation. JDBC should be used to store and retrieve complaint information from a relational database.	65 to 68
18	Courier Tracking System Develop a Courier Tracking System using Java and an appropriate GUI framework. The application should allow the user to register courier details, search for a courier using a tracking ID, view courier information, update delivery status, and delete courier records. The application should demonstrate suitable Java programming concepts, GUI components, event handling, and exception handling. JDBC should be used to store and retrieve courier information from a relational database.	69-71
19	Task Management System Develop a Task Management System using Java and an appropriate GUI framework. The application should allow users to create tasks, view tasks, search for a particular task, update task details or status, and delete completed or unwanted tasks. The application may categorize tasks based on priority or status. JDBC should be used to store and retrieve task information from a relational database.	301-303

20	Blood Donor Information System Develop a Blood Donor Information System using Java and an appropriate GUI framework. The application should allow the user to register donors, view donor information, search for donors based on suitable criteria such as blood group, update donor details, and delete donor records. The application should demonstrate Java OOP concepts, GUI components, event handling, and input validation. JDBC should be used to store and retrieve donor information from a relational database.	304,305,312
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